

International Islamic University Chittagong

Centre for General Education (CGED)

Faculty of Science

Midterm- Examination, Spring -2026

Course Code: GEEM-1101

Course Title: Text of Ethics and Morality

Marks: 30

Duration: 1.50 hours

All questions are compulsory

SL	Answer the following questions	Marks	CLOs	Blooms taxonomy domain
01	☒ a) Discuss the necessity of Ethics and morality in our professional life.	2x5=10	CLO-1	Analyze & Remember
	☒ b) Analyze the necessity of Arabic language in our country.			
	Or			
	a. "This day truthfulness shall benefit the truthful" Explain the benefits of honesty in the light of this verse.	2x5=10	CLO-1	Create
	b. Explain the lessons & teaching of surah al-Faatiha.			
02	☒ a. Explain the bad impact of bad behaviors in the society mentioned in Sura Hujrah.	3	CLO-1	Analyze & Understand
	☒ b.) Write down the stages of the creation of humankind.	4		
	☒ c. Write down the teachings of Sura al-Iklas.	3		
03	☒ a. Explain the duties of husband and wife in the family as mentioned in the holy Quran.	2x5=10	CLO-2	Remember & Create
	☒ b. State the Quranic-rules in the using of social media.			

International Islamic University Chittagong

Centre for General Education (CGED)

Mid-term Examination Spring-2026

Course Code: GEEL-1106

Course Title: Advanced English

Time: 1 Hour & 30 Minutes

Full Marks 30

Section-A: Reading Part

Read the following passage carefully and answer the questions.

Animals like the lion are known as carnivorous animals. The word 'carnivorous' is derived from Latin words meaning 'eater of flesh'. You have already read about wild lions and you know that they feed on smaller animals. But many other animals are vegetarian, and they are called herbivorous animals. In this group of animals are found all the common farm animals like the horse, the cow and the sheep. Examples of wild animals that are herbivorous are the giraffe and the elephant, instead of saying 'carnivorous animal' and 'herbivorous animal' you can say 'carnivore' and 'herbivore'. You may be surprised to learn that there are carnivorous plants as well as carnivorous animals. Plants usually obtain food from the water in the soil and from the salts that are dissolved in it. But some plants that grow in poor soil need other food to make up for the lack of food in the soil itself. To make up for this deficiency they trap small insects and digest their bodies. Carnivorous plants use ingenious devices to trap insects for their food. The pitcher plant is a common carnivorous plant in tropical forests. This plant has a clever trap shaped like a pitcher or jug. It even has a lid to keep out the rain. The mouth of the pitcher is covered with a sweet, sticky substance, like honey or nectar. Insects come to the plant to feed on this substance. When they have eaten all that is round the mouth they crawl into the pitcher to look for more. There is more honey at the bottom and they go down to feed on it. The inner wall of the pitcher is covered with fine hairs. These hairs point downwards, so that the insects can be digested and absorbed as food.

1. Answer according to the instruction given.

01×8=08

- "The pitcher plant grows well in deserts". Is the statement true or false?
- Butterfly is a/an _____ that comes out from cocoon. Fill in the gap with suitable word from the passage.
- Why does the pitcher plant trap insects? (Answer in ONE complete sentence)
- "You may be surprised to know that there are carnivorous plants." Rewrite the sentence replacing the words underlined.
- A matter or material that something is made of. Find a synonym from the passage.
- What consequences do the insects trapped in the plant face? (Answer in ONE complete sentence)
- Which can't be tamed. (Find a synonym from the passage)
- The orphan boy looked _____ before his master. (Fill in gap with word from the passage.)

Read the following passage carefully.

With 33 percent of its population aged between 18 and 35, Bangladesh faces a pressing challenge: creating enough jobs for its youth. Unfortunately, joblessness is on the rise, with around 55 lakh young people aged 15 to 24 being economically inactive. This reality is a major threat to Bangladesh's development. The consequences are visible in a surge of teenage gang violence, substance abuse, online gambling, urban crime, and gender-based harassment and

violence. A recent policy paper by the General Economics Division (GED) lays out the scale of the crisis and, crucially, how these risks are interconnected. Youth marginalization and limited access to economic opportunity are at the root of these overlapping problems. Young people aged 15 to 18 years are particularly vulnerable. An estimated 83 lakh young people are affected by substance abuse. At the same time, online gambling is expanding rapidly as idle youth look for ways to make a quick buck or kill time, many becoming addicted. Rising urban violence is another warning sign. In 2024, nearly 50 youth gangs were active in Dhaka alone, involved in drug peddling, extortion, and other crimes. Similar patterns are emerging outside the capital. The links are cyclical: gang involvement often leads to substance abuse, which in turn deepens ties to criminal networks. Breaking this cycle requires urgent and coordinated action. Otherwise, what should have been a demographic dividend could turn into a demographic burden.

Meanwhile, current interventions fall far short. Existing training programs reach only 3.64 percent of economically inactive youth, leaving the vast majority without pathways to employment. In terms of curbing drug addiction among the youth, government plans to add 1,679 rehabilitation beds will help only a fraction of the 83-lakh individuals affected by addiction.

The GED report should be treated as a wake-up call. Bangladesh must place youth development at the center of its policymaking. This means rethinking employment strategies by creating public works programs such as environmental cleanups, expanding part-time jobs for students, and scaling up vocational training. Polytechnic institutions need urgent reform, with better teachers and updated curriculum aligned with market demands.

There must also be widespread awareness campaigns to teach parents how to nurture their teenage children, to sensitize teachers and youth leaders to be mentors, and to educate young people on the consequences of substance abuse, online gambling, and criminal behavior. Young people must be made to feel valued by the government, their families, and society. Bangladesh's youth remains its greatest asset, if given the opportunity to thrive.

2. Answer according to the instruction given.

01×7=07

- Something difficult requiring effort or courage to overcome. (Find a synonym from the passage)
- A boy interested to _____ needs to be punctual. (Fill in the gap with suitable word from the passage.)
- What do you understand by substance abuse? (Answer in ONE complete sentence)
- The GED report should be treated as a wake up call. (Rewrite the sentence replacing the words underlined.)
- Easily hurt or affected physically or emotionally. (Find a synonym from the passage)
- What has been emphasized most in the passage as a big challenge for Bangladesh? (Answer in ONE complete sentence)
- Make a sentence with the phrase "employment strategies". Don't quote directly from the passage

Section B: Grammar Part

3. Answer the questions according to the directions.

1x5=5

- a. My sister with my parents (are/ is) attending the program. (Show proper sub -verb agreement)
- b. If Shefa had not mentioned her name, I hardly think I (recognize) her. (Use the right form of verb)
- c. Rafi is going to open a workshop tomorrow. (Change the voice.)
- d. Fifty liters of oil (are/is) enough to finish the project. (Show proper sub verb agreement)
- e. I want to get my documents (photocopy) clearly. (Use the right form of verb)

Section-C: Writing part

4. Answer any one of the following

05

Write a paragraph on the topic given below:

Use of Artificial Intelligence in Education

Or

Razia, like many other homemakers is facing very hardest time now for immense crisis of money. She lives in a slum with her day laborer husband and five children.. She does not know how to pay house rent and provide meals to her children. Now she.... (Expand these sentences into a complete story)

Speaking test. (Will be taken by the concerned teacher in a convenient time)

05

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Mid Term Examination, Spring-2026
Course Code: PHY-1101 Course Title: Physics-I
Total marks: 30 Time: 1 hours 30 minutes

[Answer **all** the questions. Figures in the right-hand margin indicate full marks.]

- | | | | | | |
|----|------|--|------|---|---|
| 1. | ✓ a) | Mention the perpendicular axes theorem of moment of inertia. | CLO1 | R | 1 |
| | ✓ b) | Find an expression for the moment of Inertia in the case of the perpendicular axes and parallel axes, with a proper theorem explanation and figure for a plane lamina. | CLO2 | U | 7 |
| | ✓ c) | A flywheel of mass 150 kg and radius 0.75 m makes 18 revolutions per second. Calculate (i) the rotational kinetic energy of the flywheel. Assume that the whole mass of the flywheel is concentrated at the rim. | CLO3 | A | 2 |

- | | | | | | |
|----|------|--|------|---|---|
| 2. | ✓ a) | Define Gravitational Potential Energy. | CLO1 | D | 1 |
| | ✓ b) | Derive the Gravitational Potential due to a Spherical Shell at a point outside the shell in detail with proper figure. | CLO2 | U | 6 |

OR

Prove the law of conservation of momentum for bodies under elastic collision with a proper explanation and figure. [Consider the net force on the system is zero].

- | | | | | |
|------|--|------|---|---|
| ✓ c) | A ball of mass 8.20 kg is on the launchpad at the Kennedy Space Center (before launch). Find (a) the gravitational potential energy in MJ, and (b) the total mechanical energy in MJ. (Given, the radius of the earth, $R = 6.37 \times 10^6$ m) | CLO3 | A | 3 |
|------|--|------|---|---|

- | | | | | | |
|----|------|---|------|---|---|
| 3. | ✓ a) | State Heisenberg's Uncertainty Principle. | CLO1 | D | 1 |
| | ✓ b) | Find an expression for the Schrodinger time-dependent equation. | CLO2 | U | 6 |
| | ✓ c) | Given the maximum diameter of a nucleus is 2×10^{-14} m, find (i) the minimum uncertainty in the momentum of an electron if it were to stay inside the nucleus, and (ii) if the maximum energy of an electron in the atom is 4MeV, verify that the electron exists inside the nucleus. | CLO3 | A | 3 |

OR

A rigid body is rotating about Z- axis of rotation, and the center of mass is at a distance of 4 cm from that axis of rotation. If the total mass of the object is 5 kg. Find (i) the moment of inertia, I_z , (ii) If $I_x = 0.005 \text{ kg-m}^2$, find the value of I_y .

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Mid Term Examination

Program: B. Sc. in CSE
 Course Code: MATH-1107
 Time: 1:30 hours



Semester: Spring 2026
 Course Title: Mathematics-I
 Total Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Please answer the several parts of a question sequentially.
 (iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1:	Compute the functions, limit and continuity of a function, derivatives, integrals and extrema of single-variable and/or multivariable functions.
CLO2:	Understand the techniques of differentiation and integration.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

- | | Marks | CLO | DL |
|--|-------|------|----|
| 1. a) Find the domain and range of the function, $f(x) = \sqrt{\frac{x+2}{x-2}}$ | 3 | CLO1 | R |
| b) Test the differentiability of the following function at $x = 0$
$y = f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x < 0 \\ 3 - 2x, & 0 \leq x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$ | 4 | CLO1 | U |
| c) Test the continuity of the following function at $x = 0$
$y = f(x) = \begin{cases} \frac{1 - \cos x}{x^2}; & x \neq 0 \\ 1; & x = 0 \end{cases}$ | 3 | CLO1 | U |
| Or)
d) Evaluate the limit using L'Hospitals rule, $\lim_{x \rightarrow \infty} 2^x \sin\left(\frac{a}{2x}\right)$ | | | |

2. a) Find the differential co-efficient of $\log_a x$ by the geometrical method. 3 CLO2 U
- b) Differentiate $\sec^{-1} \frac{1}{2x^2-1}$ with respect to $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ 4 CLO2 U
- Or)
- c) Find the differential coefficient of the following functions with respect to x
- (i) $y = (\sin^{-1} x)^{\ln x}$ (ii) $\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = 1$
- d) If $y = \cos^2 x + \frac{1}{\cos^2 x} + \frac{1}{\cos^2 x} + \frac{1}{\cos^2 x} + \dots \dots \dots \infty$ then find $\frac{dy}{dx}$ 3 CLO2 U
3. a) If $x = \cos \sqrt{y}$ then using the Leibnitz theorem show that,
 $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$ 4 CLO2 U
- b) Verify the Roll's theorem for the function $f(x) = e^x \sin x$ in the interval $(0, \pi)$. 3 CLO2 U
- Or)
- c) Verify Mean Value theorem for the function,
 $f(x) = x(x-1)(x-2)$ in the interval $(0, 0.5)$.
- d) Expand $\ln x$ in ascending power of $(x-2)$ by Taylor's Series. 3 CLO2 U

International Islamic University Chittagong
 Department of Computer Science & Engineering
 B.Sc. in CSE, Midterm Examination, Spring 2026
Course Code: CSE-1121 Course Title: Computer Programming I
Total Marks: 30 Time: 90 minutes

Answer all of the following questions. Figures in the right-hand margin indicate full marks, COs and DLs.
 [XYZ are the last three digits of your student ID. For example, if your ID is C253176, then X = 1, Y = 7, and Z = 6.]

- | | | | |
|---|-------|-----|----|
| | Marks | CO | DL |
| 1 (a) Why do we use long long int? Also, why do we use double when we already have float? | 2 | CO2 | C2 |
| (b) Consider the following variable declarations and initial values in a C program: | 3 | CO2 | C3 |

int a = 5, b = 10, c = 2;

Show how each of the following expressions will be calculated step by step using the precedence and associativity rules. An example has been solved for you:

a * b + c
 => a * b + c
 => 50 + c
 => 52

i) a + b * a / b - a % b

ii) (a++ > b) ? a + b * c : b + (++c)

- | | | | |
|---|---|-----|----|
| (c) Write the output of the following C code segments. If there is any error in the code, fix it. | 3 | CO1 | C3 |
|---|---|-----|----|

(i) int a = 100000;
 int b = pow(a, 2);
 printf("%lf", sqrt(b));

(ii) double num = -5.7;
 double result = abs(num);
 printf("%lf\n", result);

— OR —

- | | | | |
|--|---|-----|----|
| The following C program is intended to read a student's obtained marks and total marks, then print the percentage. The code compiles but produces wrong output. Identify the bugs and rewrite the program correctly. | 3 | CO1 | C3 |
|--|---|-----|----|

```
#include <stdio.h>
int main() {
    int obtained, total;
    scanf("%d %d", obtained, &total);
    float percentage = obtained / total * 100;
    printf("Percentage = %d%\n", percentage);
    return 0;
}
```

Sample Input: 75 100

Expected Output: Percentage = 75.00%

- | | | | |
|--|---|-----|----|
| (d) Write a C program that takes an integer representing the size of a file in Megabytes (MB). Convert this into Gigabytes (GB) and the remaining Megabytes (MB). (Note: 1 GB = 1024 MB) | 2 | CO2 | C3 |
|--|---|-----|----|

Sample Input	Sample Output
2500	2 GB, 452 MB
1024	1 GB, 0 MB
500	0 GB, 500 MB

- | | | | |
|--|---|-----|----|
| 2 (a) Find the output of the code snippet for the following initial values of a: (i) 6 (ii) 8 (iii) 3 (iv) Z | 2 | CO2 | C3 |
|--|---|-----|----|

```
int b = a / 2;
if (a > 0)
    if (a == 6)
        a = a + 3;
    else
        a = a - 2;
        b = b + 5;
printf("%d %d\n", a, b);
```

(b) The following C program should take three integers as input representing three sides of a triangle and print: 4 CO1 C3

"Not a Triangle" → if triangle inequality fails
 "Equilateral" → all sides are equal
 "Isosceles" → exactly two sides are equal
 "Scalene" → all sides are different

```
int a, b, c;
scanf("%d %d %d", &a, &b, &c);
if (a + b <= c)
    printf("Not a Triangle");
else if (a == b || b == c || a == c)
    printf("Isosceles");
else if (a == b && b == c)
    printf("Equilateral");
else
    printf("Scalene");
```

The code has **two logical errors**. Identify the errors and provide **two sets of test cases**, i.e. two different inputs for a, b, and c, where the buggy code produces **two different wrong outputs**, with a brief explanation for each. At least one of the three numbers in each test case must contain the digit **Z** (the last digit of your ID) i.e., if Z = 6, valid numbers include 6, 16, 60, 26, etc.

(c) Write a C program that takes a positive integer (number of Bits). Break it down into KB, Byte, and Bit. (1 KB = 1024 Bytes, 1 Byte = 8 Bits). Output the positive units only: "A KB, B Byte, C Bit". Don't write any zero units. 4 CO2 C3

Sample Input	Sample Output
10	1 Byte, 2 Bit
8192	1 KB
16393	2 KB, 1 Byte, 1 Bit

— OR —

Write a C program that determines the type of card a person in Bangladesh is eligible for, based on their income, occupation, and gender, according to the following criteria: 4 CO2 C3

Card Type	Income Condition	Occupation Condition	Gender Condition
Family Card	Strictly less than 1000	Unemployed	Female
Farmer Card	Strictly less than 5000	Farmer	Male
Athlete Card	No condition	Cricketer or Footballer	Both genders

Now you will be given three integers A ($0 \leq A \leq 100000$), B ($0 \leq B \leq 3$) and C ($0 \leq C \leq 1$) as input of information of a person. Here A denotes income, B denotes occupation (B=0 for Unemployed, B=1 for Farmer, B=2 for Cricketer, B=3 for Footballer) and C denotes gender (C=0 for male and C=1 for Female) of the person. Using the criteria table print "Family Card" or "Farmer Card" or "Athlete Card" if the person deserves it, otherwise print "Red Card".

Sample input	Sample Output
500 0 1	Family Card
1000 0 1	Red Card
90000 3 0	Athlete Card

3 (a) Write the output of the following code segment:

2 CO2 C3

```
int i, j = 2;
for(i = 1; i <= 10; i++) {
    if(i % 3 == 0) continue;
    if(i > 7) break;
    printf("%d %d\n", i, j);
}
```

(b) A code was written to calculate the sum of series: $1 \cdot 1^2 + 2 \cdot 2^2 + 3 \cdot 3^2 + \dots + n \cdot n^2$, where $(1 \leq n \leq 1000)$. But unfortunately the output of this code is unexpected. You are given the code, identify the problem and rewrite the right one.

3 CO2 C3

```
int i, n, sum = 1;
scanf("%d", &n);
for(i = 0; i < n; i++) {
    sum = sum + pow(i * i, 2);
}
printf("Sum = %d", sum);
```

– OR –

Alex has invented a new mathematical term called "FLONTRIX", represented using the symbol $^*!$.

3 CO2 C3

This operation is defined as the product of all even numbers from the given number down to 2.

For example, $7^*! = 6 \times 4 \times 2$ and $8^*! = 8 \times 6 \times 4 \times 2$.

If there are no even numbers to multiply, the program should output -1.

Write a C program to implement this custom FLONTRIX operation.

Sample Input	Sample Output
5	8
1	-1

(c) You are managing a community charity fund that begins with a balance of 1000 BDT. The fund operates under the following rules:

5 CO2 C4

- On Odd Days (1, 3, 5 ...), generous donors contribute to the fund, increasing the current balance by x%.
- On Even Days (2, 4, 6 ...), aid is distributed to needy families, reducing the current balance by y%.
- At the end of every day, a fixed operating cost of 50 BDT is deducted from the balance.

Write a C program that takes one integer n (total number of days) and two floating-point values x (donation percentage on odd days) and y (distribution percentage on even days) as input.

Output the final balance after n days.

However, if the balance drops to 0 or below at any point, the fund can no longer operate — it stops immediately and prints: "Fund exhausted at Day M ", where M is the day it happened.

Sample Input	Sample Output
5 10 5	931.90
10 2 50	Fund exhausted at Day 6
5 12.5 4	1032.88

Explanation of Sample 1: (initial balance = 1000.00):

- Day 1 (Odd): $1000 + (10\% \text{ donation}) - 50 \text{ cost} = 1100 - 50 = 1050.00$
- Day 2 (Even): $1050 - (5\% \text{ distributed}) - 50 \text{ cost} = 997.50 - 50 = 947.50$
- ... (continues for Days 3 and 4) ...
- Day 5 (Odd): $892.64 + (10\% \text{ donation}) - 50 \text{ cost} = 981.90 - 50 = 931.90$

Final balance = 931.90

International Islamic University Chittagong
Department of Computer Science and Engineering

Mid-term Examination, Spring-2026

Course Code: **EEE-1121**

Time: 1 hour 30 minutes

Program: **B.Sc. Engineering in CSE**

Course Title: **Basic Electrical Engineering**

Full Marks: 30

[Answer all the questions from the following. Figures in the right margin indicate full marks]

- 1) a) Discuss the concept of KVL and determine the voltages in the circuit shown in Fig:1a. CO-1 Ap 05

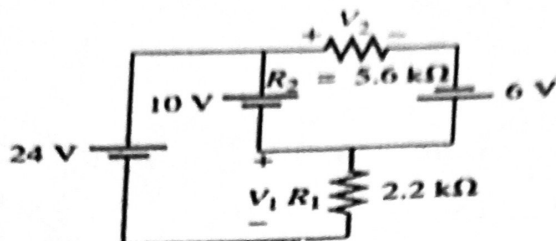


Fig: 1a

- 1) b) Find the value of v_o and i_o in the circuit of Fig:1b. CO-2 Ap 05

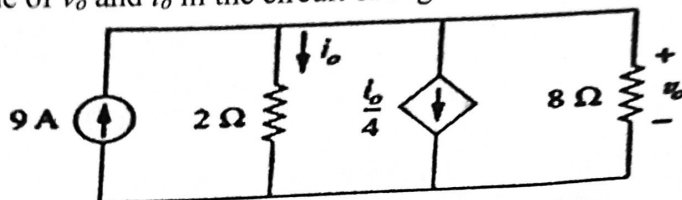


Fig: 1b

- 2) a) Using nodal analysis find the node voltages of the network of Fig:2a and also determine the current through the 4Ω resistor. CO-1 Ap 05

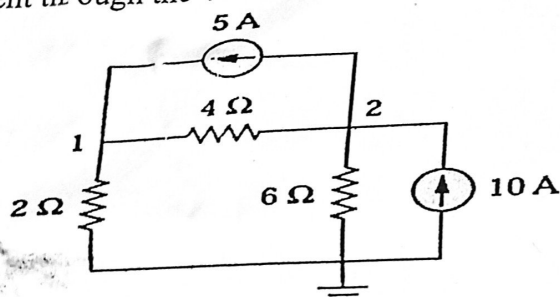


Fig: 2a

- 2) b) For the circuit below in Fig:2b, find the branch currents I_1 , I_2 , and I_3 using mesh analysis. CO-2 Ap 05

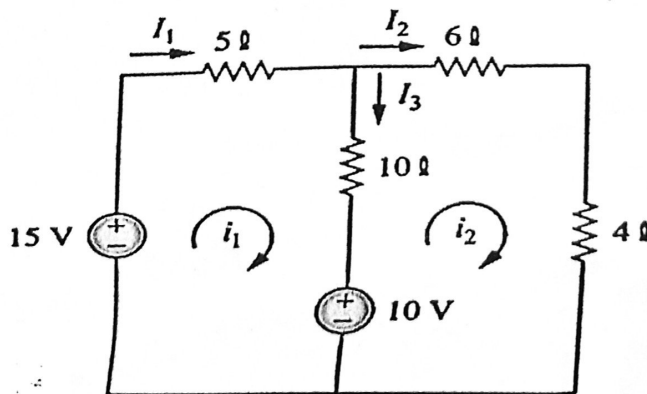


Fig: 2b

- 3) a) Apply the superposition theorem to find v in the circuit of Fig:3a

CO-2 Ap 05

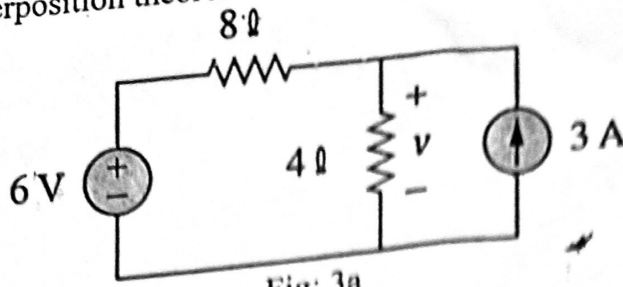


Fig: 3a

- 3) b) Find the Norton equivalent circuit of the circuit in Fig: 3b at terminals $a-b$.

CO-2 E 05

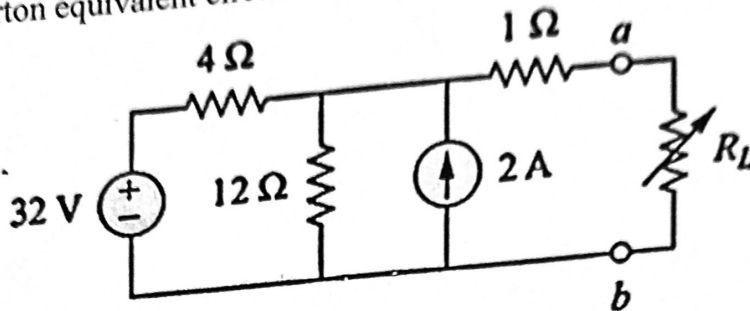


Fig: 3b

OR,

- 3) a) Use source transformation to find v_o in the circuit of Fig:3c

CO-2 Ap 05

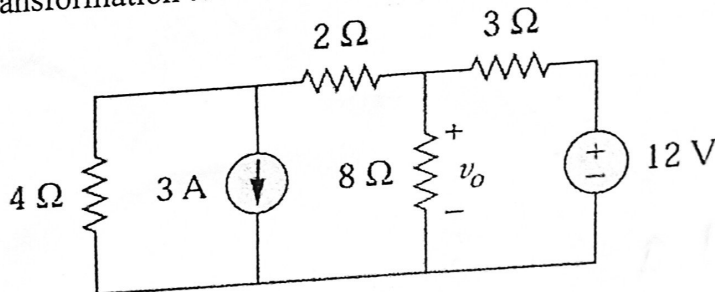


Fig: 3c

- 3) b) Find the Thevenin equivalent circuit to the left of the $a-b$ terminal of Fig:3d. and determine the current through the load resistor R_L (2Ω).

CO-2 E 05

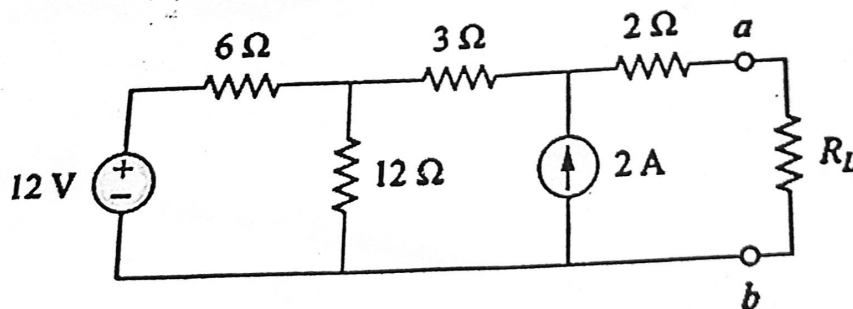


Fig: 3d